Each country's share of the cut:

$$\text{Cut}_i = \min\left(\text{Aid}_i^{2023},\ \lambda \times \text{Capacity}_i \times \frac{1}{1 + 4 \times \text{Need}_i} \times \frac{1}{1 + 4 \times \text{Efficiency}_i}\right)$$

1. Total cuts
    1. $4.7bn in cuts a year. This total stays fixed; the model only changes how it's split between countries.
2. New tax revenue
    1. Benitez et al. estimates how much tax revenue each country could realistically reach (setting a ceiling), and Akitoby et al. estimates how fast countries get there each year from 2027 to 2030. The extra revenue is then multiplied by the country's current health share of government spending.
3. Budget shifts
    1. Countries move more of their current budget into health. The increase is capped at each country's best 4-year rise in health's share of government spending, and it phases in gradually over 2027–2030.
    2. Kharas & Sarkar scored 144 countries from 0 to 100 on eight areas, including revenue, debt and governance. A low score means low fiscal space. The score sets how much of the shift a country can actually make.
        1. This isn't really what the index was made for. It measures how much extra spending a government can take on, not how flexible its current budget is. We still use it because it makes the model more conservative: countries that scored lower on the index get a smaller budget shift, and countries at or below 48.8 get none, instead of every country being assumed to match its best 4-year rise.
    3. Budget shift = Score × best 4-year rise % × total government spending × share of the shift taking place in each year.
4. Capacity = new tax revenue for health + budget shift. This is how much each country puts toward replacing lost aid.
5. Disease need
    1. Victor's disease need index adjusts the cut so high-burden countries are protected.
    2. Need discount = 1 / (1 + 4 × need). Each country's capacity is multiplied by this before the cut is calculated, so high-need countries are asked to give up less. The country with the highest need has its capacity counted at a fifth.
6. Efficiency (cost per life saved)
    1. This is meant to show where a dollar of aid has the most impact. The dashboard estimates how many lives $1M of aid saves in each country for HIV, TB, malaria and immunization. We weight those by how each country's aid is split between the four, and cost per life saved is just $1M divided by lives saved.

2. Each country's lives per $1M is divided by the highest one, so efficiency goes from 0 to 1. Niger is the top at about $730 per life saved, and the median country is about $2,240. The 17 countries that aren't in the dashboard get the average for their income group.
3. Efficiency discount = 1 / (1 + 4 × efficiency). It works the same way as the need discount, so countries where aid saves lives cheaply are asked to give up less. For example, Afghanistan's capacity goes from $30m to $5.8m once both discounts are applied.

7. Deciding who gets cut
    1. Each country's cut = λ × its capacity after both discounts, but never more than all of its aid.
    2. λ is the share of that capacity every country is asked to give up. The model keeps raising λ until the cuts add up to exactly $4,716m. Since everyone gets the same λ, every country gives up the same fraction of what it can afford (after the protections).
    3. Here λ = 36.6%. Nigeria's capacity after discounts is $36.7m, so its cut is $13.4m. If λ × capacity is more than a country's whole aid, it's cut to zero instead and λ goes up a little so everyone else covers the difference. 19 countries end up with no aid.
    4. λ is a lot higher than the 9% in the capacity-only version because the discounts shrink everyone's capacity, so λ has to be bigger to get to the same $4.7bn.
    5. A country's percentage cut depends on how many times over it could replace its own aid. Countries with almost no capacity, like Mozambique, lose almost nothing (0.1% of their aid).

Note: Why the discounts use 1 / (1 + 4 × score)

This can be changed to make the discounts more/less impactful. The exact level I chose here was somewhat arbitrary and used because it's a good middle ground and it scales nicely:

- No discount at zero: A country with a score of 0 keeps its full capacity.
- Strong protection at the top: A country with a score of 1 has its capacity counted at 1/(1 + 4), or a fifth, so it gives up 80% less than a country with the same capacity and a score of 0. A country at the top for both need and efficiency is counted at 1/25.
- Never reaches zero: Every country still takes some of the cut, so no single score can take a country out of the formula completely.
- Smooth: Each step up in score protects a little more, so there are no cliffs or tiers.

The 4 controls how strong the protection is. With 2 the top country would be counted at a third, and with 9 at a tenth. We went with 4 as a middle ground because it's strong enough to change who gets cut but capacity still matters most. Need and efficiency both use 4 so neither one outweighs the other, and both can be changed on the Assumptions tab.